

CPR Esoft Quiz 3

Question #: 1

A 48-year-old man comes to the physician because of a 2-week history of red spots on his skin. He reports easy bruising but denies fever, recent infections, or new medications. Physical examination shows multiple petechiae over the arms and legs. Laboratory studies show a platelet count of 20,000/ μ L (normal: 150,000–450,000/ μ L). Hemoglobin and white blood cell counts are within normal limits. Which of the following sets of laboratory findings is most likely to be observed in this patient?

- A. I
- B. II
- ✓C. III
- D. IV
- E. V

Rationale: Bleeding time (BT) is an indicator of platelet aggregation and function. BT is prolonged in patients with thrombocytopenia. The coagulation cascade is normal in patients with thrombocytopenia, and APTT (intrinsic cascade) and PT (extrinsic cascade) are normal in patients with low thrombocyte count.

Answer I - is observed in von Willebrand disease (Bleeding time is prolonged due to defective platelet aggregation, and APTT is prolonged due to factor VIII instability)

Answer II - is seen in hemophilia A and B (Factor VIII or Factor IX are components of the intrinsic coagulation cascade, APTT prolonged)

Answer IV - is seen in patients on warfarin or in vitamin K deficiency (Factor VII of the extrinsic pathway is most sensitive to vitamin K)

Answer V - is seen in patients with defects in primary (platelet plug) and secondary (coagulation cascade) hemostasis

Attachment:

	Bleeding time (BT)	Activated partial thromboplastin time (APTT)	Prothrombin time (PT or INR)
I	Increased	Increased	Normal
II	Normal	Increased	Normal
III	Increased	Normal	Normal
IV	Normal	Normal	Increased
V	Increased	Increased	Increased

92783.jpg

Question #: 2

A 35-year-old woman comes to the physician because of a 2-week history of easy bruising and intermittent abdominal pain. She reports no history of alcohol use. Physical examination is notable for mild jaundice. Abdominal ultrasonography shows stones in the common bile duct. Laboratory studies show:

Elevated serum alkaline phosphatase (ALP)

Elevated gamma-glutamyl transpeptidase (GGT)

Prolonged prothrombin time (PT)

A diagnosis of vitamin deficiency is made. Which of the following enzyme activities is most likely decreased in this patient as a result of this deficiency?

- ✓A. γ -glutamyl carboxylase in the liver
- B. 25-hydroxylase in the liver
- C. 1-hydroxylase in the kidney
- D. Prolyl hydroxylase
- E. Superoxide dismutase

Rationale: The vignette describes a patient with fat malabsorption due to gall stones. Vitamin K (fat soluble vitamin) deficiency may be seen in patients with biliary obstruction. Vitamin K is a coenzyme for gamma-glutamyl carboxylase which is required for the post-translational modification of glutamic acid residues in the clotting factors II, VII, IX and X.

Note that prolyl hydroxylase requires vitamin C.

25-hydroxylase and 1-hydroxylase are required for the formation of 1,25-dihydroxycholecalciferol (DHCC). This is the active hormone formed from vitamin D.

Superoxide dismutase is involved in the conversion of superoxide to H₂O₂ and oxygen.

Question #: 3

A 10-month-old boy is brought to the physician by his mother because of a 3-day history of swelling of the scalp. He also has a history of prolonged bleeding from the umbilical stump at birth. Physical examination shows swelling of the scalp and multiple bruises on his arms and legs in various stages of healing. Laboratory studies show:

Prolonged activated partial thromboplastin time (aPTT)

Normal prothrombin time (PT/INR)

Normal platelet count, bleeding time, and platelet aggregation

Which of the following is the most likely diagnosis?

- ✓A. Hemophilia A

- B. Von Willebrand disease
- C. Bernard-Soulier syndrome
- D. Vitamin K deficiency
- E. Scurvy
- F. Thrombocytopenia

Rationale: The vignette describes a patient with a bleeding disorder. Note the APTT is prolonged indicating a defect in the intrinsic pathway. Hemophilia A is an X-linked recessive disorder due to very low levels of factor VIII. Note that platelet function tests and prothrombin time (extrinsic pathway) are not affected in hemophilia A. Hemophilia B (less common than hemophilia A) may also have similar lab findings.

The PT is normal indicating no defect in the extrinsic pathway and vitamin K deficiency is unlikely.

The bleeding time, platelet count and platelet aggregation are normal. So, thrombocytopenia, Bernard Soulier syndrome (GPIB defect), and von Willebrand disease are unlikely.

In scurvy (vitamin C deficiency) there is increased bleeding due to a defect in the blood vessel wall. The PT and APTT are not affected in scurvy.

Question #: 4

A 60-year-old man is brought to the emergency department because of a 1-hour history of progressive shortness of breath, chest pain, and palpitations. Diagnostic studies confirm pulmonary embolism, and he is started on intravenous unfractionated heparin. Activated partial thromboplastin time (aPTT) is monitored to assess the effect of heparin on the intrinsic coagulation pathway. Which of the following are most likely components of this pathway?

- A. GPIb and GPIIb/IIIa
- B. von Willebrand factor and tissue factor
- C. Factors II, VII, IX and X
- ✓D. Factors VIII, IX, XI and XII
- E. Factors III, VII, X and prothrombin
- F. Factors V, X, XIII, prothrombin and fibrinogen

Rationale: The intrinsic coagulation cascade begins with Factor XII, which activates factor XI which in turn activates Factor IX. Factor IX and VIII activate factor X.

The extrinsic coagulation cascade begins with the exposure of Factor III (Tissue factor) which activates factor VII. Factor III and VII activate factor X.

The common pathway begins with the activation of Factor X either via the intrinsic or the extrinsic pathway. Factor X with Factor V activates prothrombin to form thrombin. Thrombin converts fibrinogen to fibrin. Thrombin also activates Factor XIII which stabilizes the clot and facilitates the formation of covalent cross-links in fibrin.

GPIB, GPIIb/IIIa are receptors on the platelet and are required for the formation of the platelet plug.

Factors II, VII, IX and X are vitamin K dependent clotting factors.

Question #: 5

A 26-year-old woman comes to the physician because of a 2-day history of pain, swelling, and redness in her left forearm. Three days ago, she scraped her arm on a tree branch while gardening. She has no significant medical history. Her temperature is 38.3°C (101°F). Physical examination shows a 4-cm area of erythema, warmth, and tenderness over the mid-forearm with mild edema. A small abrasion with crusted exudate is present at the center of the lesion. Which of the following mediators is most likely responsible for the pain and erythema observed at the site of injury?

- A. Leukotrienes
- ✓B. Prostaglandin E2
- C. Prostacyclin
- D. Thromboxane A2
- E. Arachidonic acid

Rationale: PGE2 is the main inflammatory mediator released by activated macrophages and is responsible for hyperemia (increased blood flow) and redness at the site of the injury. PGE2 is a vasodilator.

Prostacyclin (PGI2) is a vasodilator released by endothelial cells and it inhibits platelet aggregation to normal endothelium.

Thromboxane A2 is a vasoconstrictor released by activated platelets and facilitates platelet aggregation. Leukotrienes are mediators of an allergic reaction especially the cysteinyl leukotrienes (LTC4, LTD4 and LTE4). They are also components of SRS-A (Slow reacting substance of anaphylaxis).

Arachidonic acid (omega 6 fatty acid) is a precursor of prostanoids (prostaglandins and leukotrienes) and is NOT a mediator of inflammation.

Question #: 6

A 3-year-old girl is brought to the emergency department by her father because of a sudden onset of dry cough and shortness of breath. Her heart rate and respiratory rate are increased. Physical examination shows an anxious child using accessory muscles of respiration. Auscultation of the lungs shows bilateral expiratory wheezes. An anti-allergic medication is administered, and her symptoms rapidly improve. Which of the following best describes the pathophysiologic mechanism underlying this patient's condition?

- A. Bronchoconstriction is mediated by thromboxane A2
- B. Cyclooxygenase activity in mast cells results in synthesis of leukotrienes

- C. Slow-reacting substance-A mediates bronchodilation in the lungs
- D. Lipoxygenase activity results in the formation of PGI₂
- ✓E. Antigenic interactions in mast cells initiate cysteinyl leukotriene synthesis

Rationale: Bronchoconstriction is mainly mediated via cysteinyl leukotrienes during an allergic or anaphylactic response.

Leukotrienes are synthesized by lipoxygenase activity in mast cells during an allergic response.

SRS-A results in bronchoconstriction and results in respiratory distress during an anaphylactic response.

PGI₂ (prostacyclin) is formed from arachidonic acid by the COX (cyclooxygenase) pathway.

Mast cells release cysteinyl leukotrienes during an allergic response.

Question #: 7

A 25-year-old woman at 28 weeks' gestation is brought to the emergency department because of sudden-onset vaginal bleeding and lower abdominal cramping. Physical examination shows uterine tenderness and signs of early labor. Abdominal ultrasound confirms increased uterine activity. Analysis of uterine tissue shows increased formation of an eicosanoid that stimulates uterine contractions. Which of the following molecules is most likely responsible for this effect on the uterine tissue?

- A. Leukotriene D₄
- B. Thromboxane A₂
- C. Leukotriene B₄
- D. Leukotriene A₄
- ✓E. Prostaglandin F₂α
- F. Prostacyclin (PGI₂)

Rationale: PGF₂α is formed in uterine tissues and facilitates labor. PGE₂ also has a similar action on the uterine myometrium.

Leukotriene C₄, D₄, and E₄ are cysteinyl leukotrienes released by activated macrophages.

Leukotriene A₄ is a precursor of the cysteinyl leukotrienes.

Leukotriene B₄ is formed from LTA₄ but is less potent than the cysteinyl leukotrienes.

The endothelium releases prostacyclin. It causes vasodilation and prevents platelet aggregation.

Question #: 8

A 21-year-old woman is brought to the emergency department because of a laceration on her right hand sustained while cutting vegetables. Shortly after the injury, there is localized vasoconstriction, platelet

aggregation, and lymphocyte activation at the wound site. Which of the following eicosanoids is most likely responsible for mediating this localized response?

- ✓A. Thromboxane A₂
- B. Prostacyclin
- C. Prostaglandin E₂
- D. Prostaglandin F_{2α}
- E. Prostaglandin G₂
- F. Leukotriene D₄

Rationale: Thromboxane A₂, which is released from activated platelets, causes vasoconstriction and platelet aggregation.

Prostacyclin (PGI₂) released from the endothelium inhibits platelet adhesion to the endothelium.

Activated macrophages and neutrophils release PGE₂ and PGF_{2α} and are important local mediators of an inflammatory response (redness, swelling, and pain) at the site of injury or infection.

PGG₂ is a transient precursor of prostaglandins.

Leukotrienes are formed from arachidonic acid by the lipoxygenase enzyme and are mediators of an allergic response.

Question #: 9

A 6-year-old boy is brought to the physician by his parents because of a 4-day history of fatigue and pallor. He has been taking a sulfanilamide antibiotic for the past 5 days. Physical examination shows mild jaundice. Laboratory studies show anemia with the presence of bite cells and Heinz bodies on peripheral smear. Genetic testing shows a pathogenic variant in the glucose-6-phosphate dehydrogenase (G6PD) gene. Which of the following family members is most likely to be at risk of developing similar manifestations after exposure to a similar drug?

- A. Father
- B. Mother
- ✓C. Maternal uncle
- D. Paternal aunt
- E. Elder sister

Rationale: Sulfanilamide (sulfonamide/sulfa) antibiotics increase the production of oxidants. People who are deficient in G6PD activity are more likely to experience an adverse effect from this type of drug. G6PD is required to keep glutathione in the reduced state (GSH) which helps to protect the RBC plasma membrane from oxidative damage. Since G6PD is X-linked, it can only be given to boys from the

maternal line, so the brother of the boy's mother is also at risk.
Note that the boy's mother is a carrier of this mutation.

Question #: 10

A 4-year-old boy is brought to the physician by his parents because of a 2-day history of pallor and dark-colored urine. Physical examination shows mild jaundice and splenomegaly. Laboratory studies show anemia with evidence of increased red blood cell destruction. Further testing shows decreased activity of a metabolic pathway that impairs the red blood cell's ability to respond to oxidative stress, resulting in splenic sequestration and hemolysis. Which of the following best describes the main role of this pathway in the affected cells?

- A. Generates reactive oxygen species
- B. Generates ATP by substrate level phosphorylation
- ✓C. Generates NADPH for glutathione reductase
- D. Provides thromboxane A2 for platelet aggregation
- E. Forms 2,3-bisphosphoglycerate
- F. Generates hydrogen peroxide
- G. Generates NADPH for fatty acid synthesis

Rationale: The vignette describes a child with G6PD deficiency. G6PD is an enzyme of the pentose phosphate pathway. The pentose phosphate pathway operates to produce NADPH for glutathione reduction in the RBCs. Besides NADPH, the pentose phosphate pathway also forms Ribose (pentoses) for purine and pyrimidine nucleotide synthesis.

Reactive oxygen species are formed as a result of metabolism and during infections via NADPH oxidase, superoxide dismutase, and myeloperoxidase reactions.

Glycolysis forms ATP and 2,3-bisphosphoglycerate in RBCs.

TXA2 is formed from arachidonic acid via the cyclooxygenase pathway.

NADPH from the pentose phosphate pathway is used for fatty acid synthesis in liver cells and not in RBCs.

Question #: 11

A 16-month-old boy is brought to the physician by his mother because of a 4-day history of fever and productive cough. His medical history is notable for multiple bacterial and fungal infections over the past 4 months. Physical examination shows hepatosplenomegaly and generalized lymphadenopathy. Laboratory studies confirm impaired intracellular killing by neutrophils. The dihydrorhodamine test is negative, indicating the absence of respiratory burst activity in macrophages. Which of the following reactions is most likely directly affected in this patient?

- A. Formation of hypochlorous acid (HOCl) from H_2O_2
- B. Oxidation of glucose 6-phosphate to 6-phosphogluconate
- ✓C. Formation of superoxide from oxygen
- D. Conversion of H_2O_2 to water
- E. Formation of nitric oxide from arginine
- F. Formation of prostaglandins from arachidonic acid
- G. Conversion of adenosine to inosine

Rationale: A positive respiratory burst test is due to the activity of NADPH oxidase. Macrophages increase oxygen utilization during phagocytosis due to increased activity of NADPH oxidase.

In patients with NADPH oxidase deficiency, an increase in oxygen consumption by macrophages is NOT observed during phagocytosis (negative respiratory burst test).

The reaction catalyzed by NADPH oxidase is the conversion of oxygen to superoxide.

Review the enzymes catalyzing the listed reactions:

Formation of hypochlorous acid (HOCl) from H_2O_2 - Performed by myeloperoxidase

Oxidation of glucose 6-phosphate to 6-phosphogluconate - Performed by glucose 6-phosphate dehydrogenase (G6PD)

Formation of superoxide from oxygen - Enzyme is NADPH oxidase

Conversion of H_2O_2 to water - Enzyme is Glutathione peroxidase

Formation of nitric oxide from arginine - The Enzyme is nitric oxide synthase

Formation of prostaglandins from arachidonic acid - The Enzyme is cyclooxygenase

Conversion of adenosine to inosine - The Enzyme is adenosine deaminase (Deficiency results in SCID)

Question #: 12

A 30-year-old man presents to his physician for a routine checkup. He has no significant past medical history and reports no symptoms. Laboratory studies show normal red blood cell counts. In vitro testing of his erythrocytes shows that hydrogen peroxide generated during normal cellular metabolism is rapidly converted to water. Which of the following enzymes is most likely responsible for this reaction?

- ✓A. Glutathione peroxidase
- B. Glutathione reductase
- C. Superoxide dismutase
- D. NADPH oxidase
- E. Myeloperoxidase
- F. Cyclooxygenase-1

Rationale: Glutathione peroxidase converts hydrogen peroxide into water in the presence of reduced

glutathione. Glutathione reductase converts oxidized glutathione (GS-SG) to the reduced glutathione (GSH) using NADPH from the pentose phosphate pathway. Superoxide dismutase converts superoxide into hydrogen peroxide. The respiratory burst reaction requires NADPH oxidase in WBC, which converts oxygen to superoxide. Myeloperoxidase in WBCs forms hypochlorous acid. COX-1 acts on arachidonic acid to form the prostaglandins.

Question #: 13

An 85-year-old man volunteers to participate in a clinical study focusing on the aging of hemopoietic organs. Imaging studies and a bone marrow biopsy are required for this study. Which of the following locations would be most appropriate for a biopsy in this patient?

- ✓A. Iliac crest
- B. Carpal bones
- C. Humerus
- D. Femur
- E. Tibia

Rationale: In older adults, hemopoiesis (hematopoiesis) is maintained primarily in the vertebrae, pelvic bones (e.g., iliac bones), sternum, & ribs.

Question #: 14

A 54-year-old man comes to the physician because of a 3-month history of fatigue and weakness. Physical examination shows pale ocular and oral mucous membranes. Laboratory studies show evidence of chronic kidney disease with anemia and a low reticulocyte count. A decrease in which of the following signaling molecules most likely resulted in these laboratory findings?

- A. Interleukin 3
- B. Interleukin 7
- C. GM - Colony stimulating factor
- ✓D. Erythropoietin
- E. Thrombopoietin

Rationale: Erythropoietin (D) stimulates red blood cell production in the bone marrow. It is synthesized in the kidney, and reduced synthesis in chronic kidney disease causes anemia.

Interleukin 3 (A) is required for the differentiation of multipotent hematopoietic stem cells into myeloid

progenitor cells.

Interleukin 7 (B) is required for the differentiation of multipotent hematopoietic stem cells into lymphoid progenitor cells in the bone marrow.

GM-colony stimulating factor (C) stimulates stem cells to produce granulocytes and monocytes.

Thrombopoietin (E) stimulates the production and differentiation of megakaryocytes, which are platelet precursors.

Question #: 15

A 23-year-old man comes to the physician because of a 3-day history of fever, muscle ache, headache, a cough productive of green mucus, and severe chest discomfort. His temperature is 38°C (100°F). Physical examination shows dried oral mucosa and a "rubbing" sound at the right lung base. An x-ray of his chest shows a radiopaque collection at the right lung base. A culture of a sputum sample confirms the presence of a common flu virus. Which of the following best characterizes the prevalent cell type in this condition?

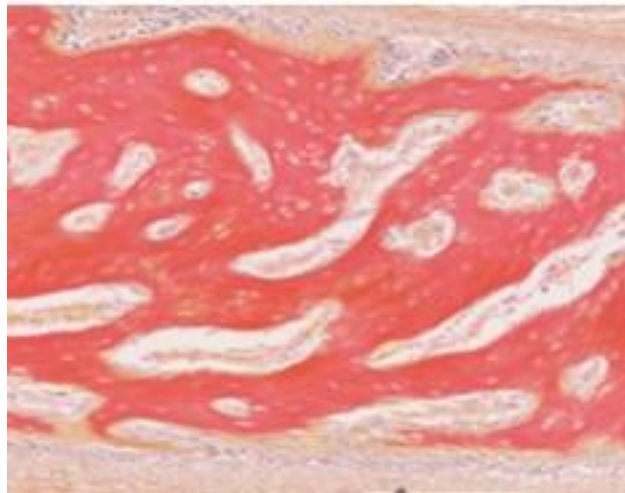
- A. Secretion of antibodies
- B. Interacts with MHC II-Ag complexes
- C. Differentiates into plasma cells
- ✓D. Interacts with MHC I-Ag complexes
- E. Matures in the bone marrow

Rationale: Because this is a virus, T-lymphocytes will be predominant, specifically the cytotoxic CD8+ T-lymphocytes. Because of the lag time of 3 days, it is more likely to be a cell mediated response of the adaptive immune arm of the system. These cells interact with MHC I-Ag complexes (D).

The Th cells which are CD4+ are the ones that interact with MHC-II Ag complexes (B), not cytotoxic T-lymphocytes .

B-lymphocytes mature in the bone marrow (E) and differentiate into plasma cells (C) which in turn secrete antibodies (A). B-lymphocytes are more characteristic of the humoral adaptive immune response.

Attachment:



trabeculae bone (mature).jpg

Question #: 16

A 25-year-old man comes to the physician because of a 2-day history of eruption of a cluster of small, red bumps on his nose and lips. Physical examination shows a yellowish crusty sore on the nose and lips. The physician suspects a bacterial infection of his skin. Which of the following best identifies a function of the cell type most likely to be prevalent in this case?

- ✓A. Secretion of antibodies
- B. Secretion of interleukin 2
- C. Secretion of interleukin 10
- D. Secretion of TNF-alpha
- E. Secretion of interferons

Rationale: B-lymphocytes are affected, leading to activation of the humoral arm of the adaptive immune system. B-lymphocytes differentiate into plasma cells that secrete antibodies. All of the other products are produced and secreted by T-lymphocytes.

Question #: 17

A 67-year-old woman comes to the physician because of a 1-week history of easy bruising and gum bleeds, and a 3-month history of fatigue. Physical examination shows areas of discoloration around her body. Laboratory studies show significantly decreased platelet count. Which of the following is most likely a precursor in the development of the decreased blood element?

- ✓A. Megakaryoblast
- B. Myelocyte
- C. Lymphoblast
- D. Reticulocyte
- E. Monoblast

Rationale: Platelets undergo the process of thrombopoiesis. Megakaryoblast is a step in this process. Myelocytes are precursors in granulocyte formation, lymphoblasts in lymphopoiesis, reticulocytes in erythropoiesis, and monoblast in monopoiesis.

Question #: 18

A 22-year-old woman comes to the physician because of a 2-month history of weakness and fatigue. Physical examination shows pallor and elevated pulse but no other abnormalities. Laboratory studies show anemia. A peripheral blood smear is shown in the attached photomicrograph. Which of the following terms best characterizes the cells of this blood smear?

- ✓A. Anisocytosis
- B. Poikilocytosis
- C. Polycythemia
- D. Spherocytosis
- E. Target cells

Rationale: The blood smear shows red blood cells of varying sizes; This is termed Anisocytosis (A). It is commonly seen in anemia.

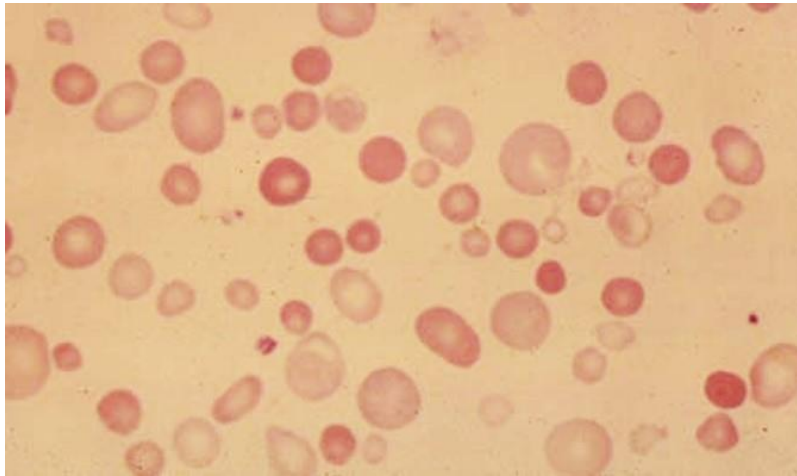
Poikilocytosis (B) means the red blood cells are of varying shapes.

Polycythemia (C) is increasing in number of red blood cells above normal limits.

Spherocytosis is the presence of spherocytes (spherical red blood cells) in the bloodstream. Red blood cells are normally described as biconcave disks with an elliptical shape.

Target cells are red blood cells that have the appearance of a shooting target with a 'bullseye'. They are seen in a variety of conditions including post-splenectomy.

Attachment:



Blood smear.jpg

Question #: 19

A 5-year-old boy is brought to the physician by his parents because of a 2-week history of abdominal pain. Stool analysis shows the presence of cysts and the ova of parasites. Laboratory studies of the blood show an increase in the count of a particular leukocyte characterized by the presence of specific granules containing major basic protein. Which of the following leukocytes are most likely elevated in this patient?

- A. Lymphocytes
- B. Neutrophils
- C. Monocytes
- D. Basophils
- ✓E. Eosinophils

Rationale: Major basic protein (MBP) is one of the proteins contained in the numerous, large, specific granules of eosinophils. MBP is cytotoxic to protozoans and helminthic parasites & accounts for the intense acidophilia of the granule.

Basophils release histamine and heparin and are elevated in allergic reactions. They have a similar structure and function to mast cells but note that they are not the same as mast cells.

Neutrophils' specific granules contain lysosomal enzymes and neutrophils function to phagocytize bacteria.

Lymphocytes lack specific granules.

Monocytes usually migrate from circulation into connective tissue where they differentiate into macrophages.

Question #: 20

A 25-year-old man is brought to the emergency department because of a 1-hour history of multiple injuries following a motor vehicle crash. Physical examination shows multiple, deep lacerations that require surgical intervention. Which of the following elements of the blood is required to maintain normal hemostasis?

- A. Lymphocytes
- B. Basophils
- C. Monocytes
- D. Neutrophils
- ✓E. Thrombocytes
- F. Eosinophils

Rationale: Platelets (thrombocytes) function in the surveillance of blood vessels, blood clot formation, and repair of injured tissue.

Question #: 21

A 7-year-old boy is brought to the physician by his parents because of frequent respiratory infections over a period of 6 months. Physical examination shows facial and cardiac abnormalities. Karyotyping shows deletion in chromosome 22. A CT scan shows an underdeveloped thymus. Which of the following immune functions will most likely be affected?

- A. Plasma cell formation
- B. Mast cell degranulation
- ✓C. Cell-mediated immunity
- D. Phagocytosis of bacteria
- E. Viral protein accumulation in dendritic cells

Rationale: Chromosome 22 gene deletion defect is seen in DiGeorge syndrome which is characterized by thymic dysfunction (aplasia). T lymphocytes which participate in cell-mediated immunity are formed in the bone marrow but normally mature and begin differentiating in the thymus. Dysfunction of the thymus therefore correlates with dysfunction in cell-mediated immunity.

Question #: 22

A 22-year-old professional athlete joins a research study for the effects of endurance exercise on the body. On his first day at exercise physiology lab, he performed high-intensity exercise drills, and his parameters are noted. Which of the following cardiovascular adaptations will most likely be reported during his endurance exercise session?

- A. Decreased RV size
- B. Decreased Diastolic filling
- C. Decreased LV size
- ✓D. Unchanged LV ejection fraction

Rationale: During endurance training, there is increased LV and RV sizes with increase in Diastolic filling. Stroke volume will be increased with no change in LV ejection fraction.

Question #: 23

A 28-year-old woman comes to the physician because of a 3-month history of mild difficulty breathing and fatigue. She has a 10-year history of obesity grade III. Her height is 5'2" and she weighs 185 pounds, and her BMI is 35 kg/m². She is advised to start a weight loss program, including dietary modification and high intensity aerobic classes five times a week for the next six months. Which of the following effects will most likely be produced by the indicated training plan?

- A. Decreased stroke volume
- B. Increased resting heart rate
- C. Increased resting arterial blood pressure
- D. Decreased ventricular muscle mass
- ✓E. Increased capillary capacity for blood flow

Rationale: Skeletal muscle blood flow capacity (BFC) is increased by exercise training due to structural vascular remodeling (in the form of angiogenesis of capillaries and remodeling of the arterial tree within skeletal muscle) and/or altered control of vascular resistance. Effects of exercise training include Increased capillary capacity for blood flow, muscle hypertrophy, increased stroke volume, reduced heart rate and resting arterial BP.

Question #: 24

A 25-year-old woman is brought to the emergency department after having been found unresponsive by her roommate in the bathtub. Examination shows deep lacerations to each of her wrists, and she is pale and unconscious. Her systolic blood pressure is 90 mm Hg. An increase in which of the following would most likely be a direct consequence of her condition?

- A. Cardiac output
- B. Atrial natriuretic peptide secretion
- ✓C. Antidiuretic hormone (vasopressin) secretion
- D. Renal blood flow

E. Venous compliance

Rationale: Hemorrhagic shock creates a condition of inadequate tissue perfusion resulting in cellular demise due insufficient oxygen flux. Physiological responses are aimed at vasopressin release, RAAS activation and catecholamine release. This leads to vasoconstriction, increased blood volume, and cardiac stimulation.

Question #: 25

A 62-year-old man is brought to the intensive care unit because of overwhelming sepsis. The patient has received 4 L of normal saline bolus fluids. Empirical antibiotics were begun with no improvement in his condition. His blood pressure is 60/30 mm Hg, pulse is 112/min, temperature is 40.6° C (105° F), and respiratory rate is 23/min. Which of the changes listed in the attached table are most likely in this patient before he received treatment?

- A. A
- B. B
- C. C
- D. D
- ✓E. E
- F. F
- G. G

Rationale: Sepsis -> vasodilation -> reduced TPR, reduced preload, reduced stroke volume. The key compensation for reduced cardiac output is to increase cardiac contractility.

Attachment:

Physiologic variable	Preload (R)	Preload (L)	Pump function	Afterload	Tissue Perfusion	Treatment
Clinical Measurement	RAP/CVP	PCWP/LVEDP	Cardiac output/index	SVR/ TPR	MvO ₂	
A	↓	↓↓	↓	↑	↓	IV fluids
B	↑	↑	↓	↑	↓	Inotropes, diuresis
C	↑	↓	↓	↑	↓	Relieve obstruction
D	↑	↑	↓	↑	↓	Relieve obstruction
E	↓	↓	↑	↓	↑	IV Fluids, Pressors
F	↓	↓	↑	↓	↑	IV Fluids, Pressors
G	↓	↓	↓	↓	↑	IV Fluids, Pressors

SHOCK TABLE.jpg

Question #: 26

A 70-year-old man comes to the physician because of shortness of breath, severe fatigue and swelling of ankles for the past 3 months. Past medical history shows several anginal attacks. His vital signs are arterial blood pressure is 185/95mmHg, heart rate is 70/minute, and respirations are 17/minute. Physical examination shows mild cyanosis. The chest x-ray shows cardiomegaly and ECG shows left axis deviation. He is prescribed a drug which causes a reduction in aortic systolic pressure. Which one of the following mechanisms of action would this drug use?

- A. a decrease in arterial compliance
- B. a decrease in venous compliance
- ✓C. a decrease in blood volume
- D. an increase in systemic vascular resistance
- E. an increase in myocardial contractility

Rationale: Answer Key: C

Correct Feedback: $MAP = CO \times TPR$, Increased systemic vascular resistance (TPR) and increased myocardial contractility (increases CO) increase MAP. $\Delta P = \text{flow} \times \text{resistance}$; A decreased arterial and venous compliance (increased arterial/ venous resistance) also increases Pressure. Low blood volume leads to reduced venous return and reduced cardiac output which leads to reduced MAP. Diuretics reduce blood volume.

Question #: 27

A 35-year-old man comes to the primary care physician with a 2-week history of progressive shortness of breath that occurs with activity. He previously exercised regularly and has never had symptoms like this before. Now he can walk only one block before becoming symptomatic. He is a non-smoker with no family history of cardiac disease. He recalls having an upper respiratory infection about a month ago that improved on its own. Physical examination shows crackles in his lungs bilaterally and an S3 gallop. An x-ray of the chest reveals cardiomegaly. He is prescribed a diuretic. What is the most likely mechanism of action of drug given to this patient for their heart failure?

- A. Increased arterial blood pressure
- B. Increased afterload
- ✓C. Increased Na^+ excretion
- D. Increased sympathetic outflow
- E. Increased preload

Rationale: Many therapeutic agents used in the management of heart failure modulate the Neurohumoral systems that are activated by compromised cardiac function. The renin-Angiotensin –

aldosterone system can be inhibited by (1) β -adrenergic antagonists, which inhibit Renin release by the juxtaglomerular cells of the kidney; (2) ACE inhibitors, which prevent the conversion of angiotensin I to the active hormone angiotensin II; and (3) spironolactone, which competitively antagonizes aldosterone binding to the Mineralocorticoid receptor. Diuretics promote Na^+ excretion and thereby counteract the Na^+ retention stimulated by activation of the renin- Angiotensin –Aldosterone system. Venodilators counteract the effect of intravascular volume expansion by increasing peripheral venous capacitance and thereby decreasing preload. Direct arterial vasodilators alleviate the α -adrenergic receptor -mediated and Angiotensin II receptor -mediated vasoconstriction induced by increased sympathetic outflow. Cardiac glycosides, β -adrenergic agonists,

Question #: 28

A 58-year-old man comes to the physician because of a 3-day history of severe headaches. He is hypertensive with hyperlipidemia. His blood pressure is 172/108 mm Hg but is reduced back down to normal after taking his medication. Administration of this drug is most likely to cause which of the following hormonal changes?

- A. Increased level of plasma renin
- B. Increased levels of aldosterone
- ✓C. Decreased formation of angiotensin II
- D. Decreased level of atrial natriuretic peptide
- E. Increased levels of antidiuretic hormone

Rationale: Angiotensin converting enzyme inhibitors (ACE inhibitors) inhibit the activity of the enzyme ACE, which decreases the production of angiotensin II. Angiotensin II is a potent vasoconstrictor. Inhibition leads to vasodilation and reduced blood pressures.

Question #: 29

A 25-year-old man is brought to the emergency department after suffering a head trauma in a car crash. His pulse is 57/min, blood pressure is 160/98 mmHg, and respirations are 18/min. Neurologic examination shows left hemiparesis and right anisocoria. CT scan of the head shows a frontal epidural hematoma. During the neurosurgical procedure to remove the hematoma, a catheter was placed in the left ventricle to measure intracranial pressure and estimate cerebral perfusion pressure and cerebral blood flow. The physician noticed many changes in the perfusion pressure in the range of 50 to 125 mmHg. However, cerebral blood flow did not change. Which of the following mechanisms best explains the constant cerebral blood flow in this patient?

- A. Arterial constriction
- B. Venoconstriction

- C. Reactive hyperemia
- D. Metabolic hyperemia
- ✓E. Myogenic response

Rationale: This centers on local control of blood flow to the brain where blood flow is kept constant over a range of pressures. The normal range of autoregulation is between ~50 to ~150 mmHg and occurs at the levels of the arterioles leading into the specific organ. It is an intrinsic response of the vascular smooth muscle in the arterioles for that organ (myogenic response, or Bowditch effect). The pressure is controlled by altering diameter or resistance (see Darcy's and Poiseuille's laws for Flow and resistance respectively). Increased perfusion or incoming pressure stretches the vasc SM and results in vasoconstriction, and reduced blood flow. Decreased perfusion pressure results in less stretch of the vasc SM and thus vasodilation.

This autoregulation occurs in tissues where volume of blood is important- the kidneys (filtration), the brain (there is limited volume in the skull). Autoregulation also occurs in coronary circulation. The myogenic response, or the Bowditch effect.

Explanation of other options:

A & B: Arteriolar constriction and venoconstriction regulate the mean arterial pressures.

C & D: Hyperemias are due to release of local vasodilator metabolites. This is not the case with experimental changes in perfusion, and there is no ischemia involved as the perfusion pressures do not go to zero.

Question #: 30

A 5-year-old girl is brought to the emergency room after falling through the ice while skating on an outdoor pond. She is removed from the water within 1 minute, but dry clothing was not available, and she is still cold and wet 20 minutes later. Which of the following mechanisms most likely helped to maintain the patient's core temperature during the period following her rescue?

- A. Cutaneous vasodilation
- B. Diving response
- C. Increased thermoregulatory set point
- D. Release of endogenous pyrogen
- ✓E. Shivering

Rationale: Shivering is a fundamental physiological response that occurs in skeletal muscles to produce heat when it is cold or during the development of fever. This response is initiated and maintained by the central nervous system.